1 Telemetry & Physical Inputs

Controller Telemetry

100 Hz TSN deterministic sync

- Motor current: $i_q(t)$ (A)
- Angular velocity: $\omega_m(t)$ (rad/s)
- Table position: $x(t)$ (m)
- Direct drive torque: $\tau_m(t)$ (Nm)
- Nut surface temp.: $T_s(t)$ (°C)

BSFD Digital Twin

Coupled electromechanical ODEs

- PMSM dq-frame electromechanics
- LuGre pre-sliding friction bristle
- Torsional-axial compliance
- Moving thermal nodal capacity
- State vector: $\mathbf{x}^{(7)^\top} \in \mathbf{R}^{11}$

Physical Specifications

PMI C3 Ground Screw (R32-10T3)

- Screw diameter: $d_0 = 32$ mm
- Lead pitch: $\ell_p = 10$ mm
- Dynamic capacity: $C_a = 21.2$ kN
- Preload force: $F_{pr} = 1.06$ kN
- Basic rated life: $L_{10h} = 10,000$ h

Degradation Thresholds

Fatigue wear boundary conditions

- Pristine boundary: $D(0) = 0.00$
- Critical spalling: $D_{\text{crit}} = 1.00$
- Early-warning threshold: $D_{\text{warn}} = 0.80$
- Rate law: ISO 3408-5 fatigue backbone
- Acceleration factor: $A_F \approx 20$

100 Hz TSN

Twin ODEs

Physics Prior

2 Offline Hierarchical Calibration & Learning

A. Deterministic Preprocessing

- Reversal-interval kinematic windowing
- Phase-synchronous median cycle isolation
- Outlier rejection & low-pass decimation

B. Kinematic Telemetry ($J = 3$)

- Motor current: $y_1 = i_q$ (A)
- Angular speed: $y_2 = \omega_m$ (rad/s)
- Torque innovation: $y_3 = \tau_{\text{load}} - \tau_{\text{nom}}$

C. Twin Parallel Simulation (6D)

- Nominal torque: $\hat{\tau}_{\text{drive}} = J_{\text{eq}}\ddot{\theta} + T_f(\dot{\theta})$
- Thermal growth: $\Delta\hat{x}_{\text{th}} = \alpha_L L_{\text{eff}}(\hat{T}_s - T_0)$
- Friction state: $z(t)$ dynamic LuGre bristle

D. Physics Residuals (4D)

- Torque discrepancy: $r_\tau = \tau_{\text{load}} - \hat{\tau}_m$
- Temperature discrepancy: $r_T = T_{\text{nut}} - \hat{T}_{\text{nut}}$
- Dissipated frictional power: $P_{\text{fric}} = \tau_f \cdot \omega_m$
- Kinematic reversal frequency: $f_{\text{rev}} = N_{\text{rev}}/T_{\text{cyc}}$

E. Multi-Channel Observation Vector: $\mathbf{y}(t_k) = \mathbf{H}\mathbf{x}(t_k) + \gamma B(t_k) + \epsilon_k \in \mathbf{R}^3$

Integrates kinematic telemetry, baseline mechanical loads, and multi-channel observation innovations

F. KI-SSDM Hierarchical Training Architecture

1. Physics Fatigue Backbone:

$$\frac{\mathrm{d}D}{\mathrm{d}z} = C_0\left(\frac{F_{\text{eq}}(z)}{C_a}\right)^p a_{\text{id}}(z)$$

Frailty: $\log C_a \sim \mathcal{N}(\mu_C, \sigma_C^2)$

2. Reversible OU Nuisance:

$\mathrm{d}B(t) = -\theta_B B(t)\mathrm{d}t + \sigma_B \mathrm{d}W_B(t)$

$\mathrm{Var}(B) = 1$, $\gamma \in [-2, 2]^3$

3. Exact Kalman Marginalization:

$\mathbf{e}_k = \mathbf{y}_k - \hat{\mathbf{y}}_k$, $F_k = \mathbf{H}_B \mathbf{P}_k \mathbf{H}_B^\top + \mathbf{R}$

Decouples $B(t)$ in closed form

4. Smooth Tanh Rate Limiter:

$\rho(\phi) = b\tanh(\mathbf{W}_2\tanh(\mathbf{W}_1\phi))_+$

Rate bounded: $e^{-b} \le \exp(\rho) \le e^b$ ($b = 0.30$)

5. Validation Linear Pool:

$p = (1-w)p_{\text{phys}} + wp_{\text{disc}}$

$w \in [0, 1]$ via held-out likelihood

6. Projected BFGS Optimization:

Stationarity: $\|\nabla_\theta \ell^*\|_\infty < 5 \times 10^{-4}$

Guaranteed monotone wear clock: $\dot{D} \ge 0$

G. Multi-Unit Identifiability

- Strict identifiability: $\mathrm{Var}(B) = 1$ anchor
- Quadrature: 25-node Gauss–Hermite integration
- Profiled likelihood over frailty prior (μ_C, σ_C)

H. Split-Conformal Calibration

- Held-out fleet calibration $\mathscr{D}_{\text{cal}}$ (10 units)
- Nonconformity score: $s_i = |\text{RUL}_i - \widehat{\text{RUL}}_i|$
- Origin-stratified empirical quantile $\hat{q}_\alpha$
- Guaranteed marginal coverage: $\ge$ **90%**

Dispatch $\theta^* = \{\hat{\theta}, \mathbf{W}, w, \kappa\}$

Calibration $\hat{q}_\alpha$

3 Online Dual State Inference

A. Live Industrial Telemetry

- Real-time ingestion: **100** Hz TSN sync
- Sampling hop: 10 ms cycle time
- Synchronous plant state $\mathbf{x}^{\text{DT}}$ tracking
- Non-invasive: zero extra sensors needed

B. Dual State Estimation

- Reversible nuisance: Kalman filtering $B(t)$
- Damage update: $D_k = D_{k-1} + \Delta D_k$
- Gamma process increments: $\Delta D_k \ge 0$
- Strictly monotone wear clock: $\dot{D} \ge 0$

C. Prognostic Engine

- Planned & probabilistic future duty cycles
- Forward Monte Carlo: 512 path projections
- Critical first hitting time criterion:
  $\text{RUL} = \inf\{t > 0 : D(t) \ge D_{\text{crit}} = 1.0\}$

D. Conformal Uncertainty

- Calibrated band: $[\widehat{\text{RUL}} - \hat{q}_\alpha, \widehat{\text{RUL}} + \hat{q}_\alpha]$
- Empirical coverage: **93.3%** / **92.5%**
- Interval sharpness: **82.5** h (3× vs LSTM)
- Finite-sample coverage guarantee: $\ge$ 90%

4 Prognostic & Decision Outputs

Latent Damage State

$\widehat{D}(t) \in [0, 1.0]$

- Monotone wear: $\widehat{D}(t)$
- Spalling limit: $D_{\text{crit}} = 1.0$
- Safety buffer at $D = 0.80$
- Physics-consistent kinetics

Predictive RUL

Point Forecast $\widehat{\text{RUL}}$

- Planned MAE: **16.40** h (4.4%)
- Probabilistic MAE: **18.29** h
- Evaluated across 30 fleets
- 6 operating wear scenarios

Calibrated Bounds

90% Conformal Band

- $[\text{RUL}_{lo}, \text{RUL}_{hi}]$
- Origin-stratified $\hat{q}_\alpha$
- Marginal coverage $\ge$ 90%
- Finite-sample exactness

Decision Support

Optimal Replacement $\tau^*(\alpha)$

- Schedule: $\tau^* = \max(0, \widehat{\text{RUL}} - \hat{q})$
- Risk budget: $\alpha = 0.10$
- Fleet failure rate: $\lesssim 3.9\%$
- Savings: **34.7%** vs periodic

Signals & Physical Parameters:

- u, i, τ, P, ω, x: Motor voltage, current, torque, power, speed, pos.
- $D(t) \in [0, 1.0]$: Irreversible fatigue damage ($D_{\text{crit}} = 1.0$)
- $B(t)$: Latent reversible Ornstein–Uhlenbeck thermal nuisance state
- Critical Failure: First hitting time $\inf\{t > 0 : D(t) \ge D_{\text{crit}}\}$

Framework Components:

- **KI-SSDM**: Knowledge-Informed State-Space Degradation Model
- **BSFD**: Precision ball-screw feed drive electromechanical twin
- Telemetry: $J = 3$ physical channels (q-current, speed, torque innovation)
- **GLS**: Closed-form Generalized Least Squares nuisance marginalization
- $\theta^* = \{\hat{\theta}, \mathbf{W}, w, \kappa\}$: Optimized hybrid model parameters

Methodological Innovations & UQ:

- Smooth Tanh rate limiter: $e^{-b} \le \exp(\rho) \le e^b$ ($b = 0.30$)
- Linear stacking: $p = (1-w)p_{\text{phys}} + wp_{\text{disc}}$ ($w \in [0, 1]$ via CRPS)
- Split-conformal calibration quantile $\hat{q}_\alpha$ on held-out units
- Distribution-free finite-sample coverage guarantee: $\ge$ **90%**

Flow Semantics & Decision Metrics:

⟶ Solid: Online inference & physical telemetry flow

- - → Dashed: Offline diagnostics, calibration & decision outputs

- Planned Duty MAE: **16.40 h**; Probabilistic MAE: **18.29 h**
- Decision Support: **34.7%** cost reduction over periodic overhaul